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phase 2 review 1.pdf

UE23CS320B- Project Phase - 2

Capstone Project - Phase 2 - Review 1

Project Title:

Efficient Contextualized History Orchestrator [E.C.H.O]

Project ID: 92

Project Guide: Dr. M. Shalima Sulthana

Project Team:

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Agenda

- **Problem Statement**
- **Abstract and Scope**
- **Suggestions from Phase-1**
- **Requirements Specification**
- **Data set and requirements**
- **Design Approach**
- **Design Constraints, Assumptions & Dependencies**
- **Proposed System / Approach**
- **Architecture**
- **Design Description**
- **Project Progress**
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The Fragmented Digital Self

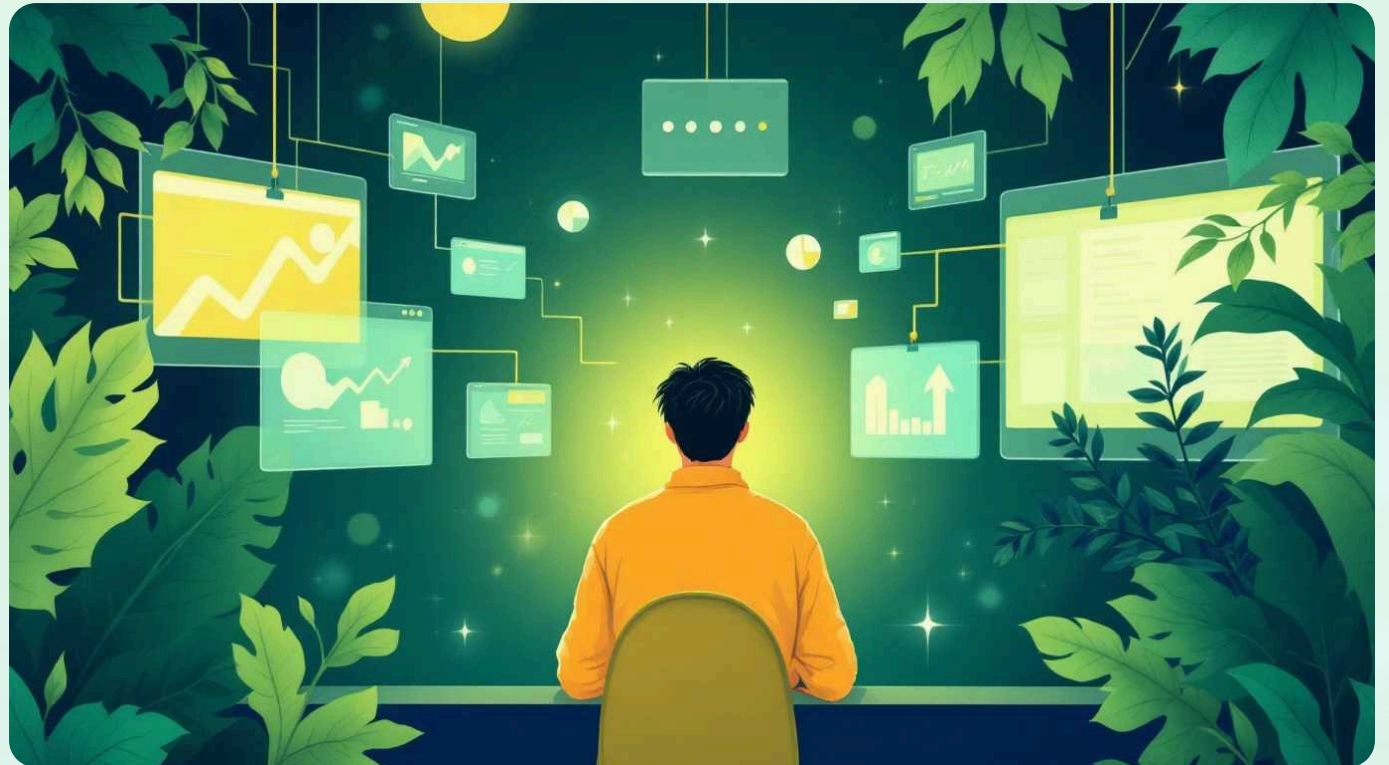
In our modern world, we generate vast volumes of digital activity daily across various platforms—emails, web browsing, video platforms. This information, crucial for studying, working, or preparing for key life events, is scattered and disconnected. This fragmentation makes it nearly impossible to recall essential details, understand relationships between resources, or reconstruct our digital journey.

Problem Statement

Existing tools offer only isolated searches, lacking awareness of user intent, cross-source context, or engagement depth. This leaves users struggling to:

- Reconstruct meaningful learning or work histories.
- Perform unified searches across all sources.
- Accurately reflect on time and attention usage.

Furthermore, most solutions rely on cloud storage, raising significant privacy and control concerns due to opaque data collection practices.



INTRODUCING

Echo: Your Local-First Personal Memory System

Echo is designed to capture, organize, and unify your digital interactions into a coherent, privacy-preserving personal memory. By operating entirely on your device, Echo empowers you to take control of your digital history without surveillance or judgment.



Privacy-by-Design

Local-first architecture ensures your data stays on your device.



Unified Semantic Search

Find anything across emails, web, and videos with intelligent retrieval.



Intent-Aware Capture

Only meaningful interactions are processed, reducing noise.

Abstract and Scope

Echo unifies fragmented digital traces, enabling you to reconstruct learning or work histories, locate relevant resources, and understand your time and attention distribution.

Abstract

Echo organizes, searches, and helps you reflect on your digital activities from Gmail, Chrome, and YouTube. It uses embeddings and retrieval-augmented generation for cross-source semantic search, all while prioritizing privacy with on-device operation.

Scope

- Meaningful activity capture from Gmail, Chrome, YouTube (desktop).
- Intent-based filtering for relevant data.
- Local storage of processed memories and semantic embeddings.
- Unified cross-source search with LLM synthesis.
- Time analytics, session tracking, and wellbeing reflection.
- Comprehensive user control over data.

Suggestions from Phase-1

Panel Suggestions – Phase I

1. **Appreciation of Concept & Real-World Relevance**

The panel acknowledged Echo as a unique and innovative idea addressing a genuine real-world problem — digital memory fragmentation across Gmail, Chrome, and YouTube. The cross-platform unification approach was appreciated for its originality and practical relevance.

2. **Full Scope Implementation Commitment**

The panel emphasized the importance of implementing the complete feature set outlined in Phase I scope, rather than delivering a reduced scope or prototype.

Response & Alignment

1. **Architectural Commitment to Full Vision**

We have structured Echo's system architecture modularly to ensure that all proposed features — Unified Search, Cross-Source Retrieval (RAG), Intent-Based Filtering, and Effort-Aware Ranking — are technically integrated into the design blueprint from the start.

2. **Scope Integrity Assurance**

The team's design have been aligned strictly with the declared scope to guarantee full implementation of all committed components rather than a partial prototype.

Requirements Specification

Functional Requirements

- Capture from Gmail, Chrome, YouTube.
- Intent-based filtering.
- Unified Search across sources
- Local structured memory storage.
- Semantic embeddings for retrieval.
- Effort-aware ranking.
- Natural language search.
- Time analytics and session computation.
- User-defined grouping and regret marking.
- Deletion, export, and privacy controls.

Non-Functional Requirements

Performance: Query <2s, Page <100ms, Dashboard <3s.

Privacy & Security: Local-first, no incognito/typed text tracking, encrypted tokens.

Reliability: FAISS rebuildable, Redis failsafe, crash recovery.

Usability: Intuitive dashboard, transparent logic, clear feedback.

Technology Stack and Justification

Component	Technology	Why Chosen
Main Database	PostgreSQL	Reliable relational storage, indexing, JSON support
Vector Search	FAISS	Fast in-memory similarity search, rebuildable index
Cache	Redis	Lightweight, TTL-based temporary storage
Backend	Python + FastAPI	High performance, async support, clean API design
Embeddings	Sentence-BERT (MiniLM)	Lightweight (384-dim), strong semantic similarity
LLM	Claude / Ollama	High-quality synthesis with optional local fallback
Frontend	React	Component-based, scalable UI
Browser Capture	Chrome Extension (Manifest V3)	Native browser activity tracking

Data set and requirements

Dataset Exploration

Source Data(YouTube, Chrome): Real-time ingestion of Chrome browsing activity (URLs, page content), and YouTube watch history (video metadata and timestamps are available).

Gmail Data: Retrieved via secure Gmail API integration, including email metadata (sender, subject, timestamps), message body content, and thread context for semantic processing and retrieval.

Data Type: Structured metadata + cleaned primary text + semantic vector embeddings (generated post-processing).

Scale: Architected to support long retention (up to two years) of unified user activity data while maintaining efficient retrieval and minimal performance degradation.

Data Requirements

Unified Format: All ingested data is stored in a Unified Canonical Memory Structure to support cross-source retrieval and temporal analytics.

Standardization: Resource URLs canonicalization (removing tracking parameters), timestamp normalization, and deduplication.

Security Processing: Local-first encrypted storage with controlled access mechanisms; designed to align with privacy-first principles.

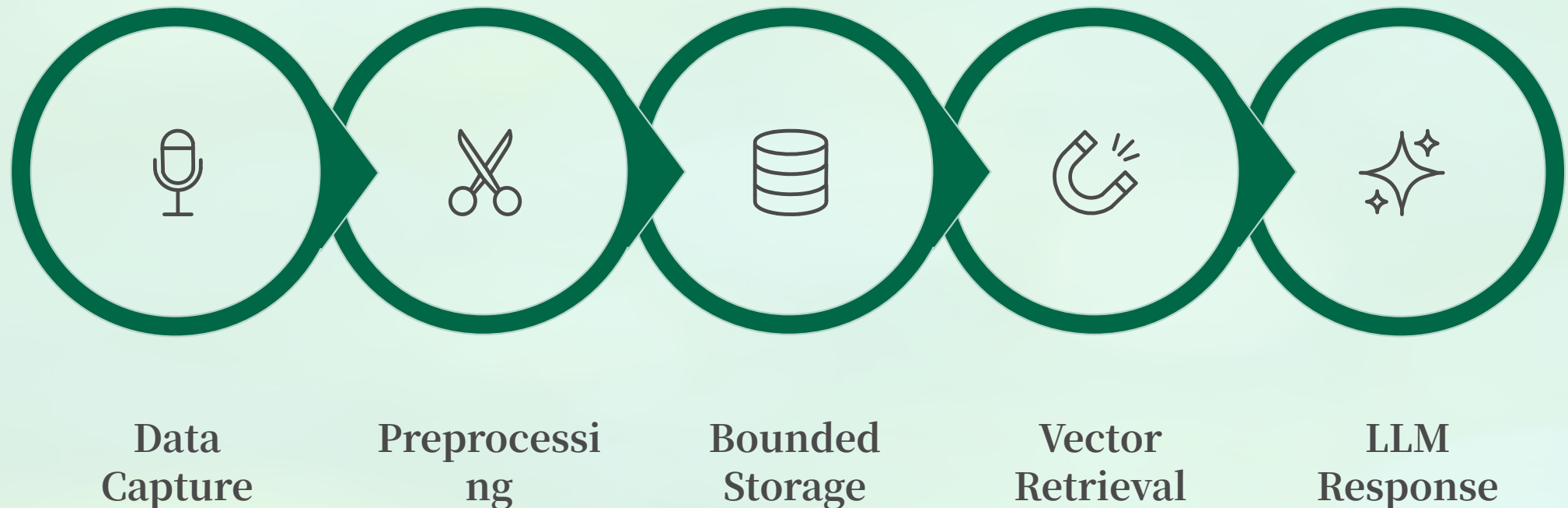
NLP Prerequisite: Extracted text is converted into semantic embeddings for deterministic RAG-based retrieval (LLM used only for synthesis, not retrieval).

Comprehensive Database Entity Catalog

Table Name	Entity Type	Category	Purpose
system_groups	Strong ▾	Core ▾	Stores fixed system categories (work, study, entertainment, personal, misc)
memory_items	Strong ▾	Core ▾	Central memory abstraction for Gmail, Chrome, YouTube
gmail_metadata	Weak Subtype ▾	Source ▾	Gmail-specific fields
gmail_attachments	Weak ▾	Source ▾	Email attachments
chrome_metadata	Weak Subtype ▾	Source ▾	Browser activity metadata
youtube_metadata	Weak Subtype ▾	Source ▾	Video metadata
memory_engagement	Weak 1:1 ▾	Behavior ▾	Tracks interaction statistics
regret_events	Weak 1:N ▾	Behavior ▾	Stores regret history

Table Name	Entity Type	Category	Purpose
user_groups	Strong ▾	Organization ▾	Stores user-created groups with soft-delete + timestamps
memory_user_group_s	Associative ▾	Organization ▾	M:N mapping between memory items and user groups
embedding_index	Weak (1:1) ▾	ML Infrastructure ▾	Stores embedding version, dimension, and index state
sessions	Strong ▾	Analytics ▾	Tracks activity sessions and dominant system group
session_memory_map	Associative ▾	Analytics ▾	M:N mapping between sessions and memory items
group_rules	Weak ▾	Automation ▾	Stores persistent classification rules
group_suggestions	Weak ▾	Automation ▾	Logs AI group suggestions + human decisions

Design Approach: A Privacy-Preserving, Intent-Aware, Modular Memory Architecture Using Deterministic Retrieval-Augmented Generation.



Echo captures meaningful user interactions, preprocesses them into structured memories, stores data locally, and enables cross-source semantic retrieval using embeddings and deterministic ranking. LLMs are used only for answer synthesis, not for decision-making or storage.

Why This Approach?

Preserves privacy, reduces noise, unifies search, ensures explainable retrieval, and maintains modularity. Balances accuracy, efficiency, and ethical considerations.

Modules:

7 functional modules organized for parallel development, clear ownership, and loose coupling.

Module	Name	Core Responsibilities
GMC	Gmail Connector	Gmail API (OAuth, webhooks), email ingestion, engagement tracking, thread context, attachment metadata
CHC	Chrome Connector	Chrome extension, intent gate logic, revisit detection (Redis), document vs app classification, time tracking
YTC	YouTube Connector	URL-based video detection, playback state tracking, watch time (foreground only), YouTube Data API, Shorts classification
STE	Storage Engine	PostgreSQL CRUD, FAISS index management, Redis cache, ID mapping, index rebuilding, transaction management
ENP	Enrichment Pipeline	Text cleaning, content extraction (Readability), system group classification (4-stage), embedding generation, transcript fetching, attachment processing
RSE	Retrieval & Synthesis	Query parsing, PostgreSQL pre-filter, FAISS semantic search, fallback chain, effort-based re-ranking, LLM context assembly, answer synthesis
WBA	Wellbeing Analytics	Time aggregation, session computation, user group management, hybrid KNN+rule classifier, weekly review queue, regret system, pattern detection, LLM insights
UIL	UI Layer	React dashboard, Chrome extension popup, FastAPI gateway, search UI, analytics/heatmap views, group management UI, settings

Benefits and Drawbacks

Benefits

- **Privacy-Preserving:** Local-first design keeps sensitive data on device, reducing external exposure.
- **High Retrieval Accuracy:** Combines semantic similarity with engagement signals for meaningful results.
- **Explainable Results:** Deterministic ranking ensures transparent and understandable outputs.
- **Modular Architecture:** Independent components make the system easy to maintain, upgrade, and extend.
- **Fast Performance:** Vector indexing and caching enable low-latency search and response times.

Drawback(s)

- **Tight Coupling Within Monolith**
- **Limited Horizontal Scalability**
- **Deterministic Pipeline Limits Autonomy**
- **Desktop Dependency**

Alternative Approaches Considered & Rejected

We thoroughly evaluated several alternative design paradigms, but ultimately rejected them in favor of Echo's unique approach due to critical limitations in privacy, accuracy, or predictability.



Cloud-Centric Logging

Easier sync, but rejected for severe privacy risks and data sensitivity.



Rule-Based Keyword Search

Simple, but rejected for poor semantic understanding and low retrieval quality.



Agentic AI Retrieval

Flexible, but rejected for unpredictable and non-explainable outcomes.



Full Surveillance Capture

Maximum recall, but rejected for excessive noise and privacy concerns.

Design Constraints, Assumptions & Dependencies

Design Constraints

- Operate entirely local (no cloud).
- No capture of sensitive data (typed text, credentials, incognito).
- Avoid background surveillance; foreground, intent-driven only.
- Bounded storage growth through retention/summarization.
- Deterministic, explainable retrieval.
- Comply with platform restrictions (browser extensions, APIs).

Assumptions

- Users primarily on desktop/laptops for study/work/entertainment.
- Gmail, Chrome, YouTube cover most relevant digital interactions.
- Engagement signals approximate user effort.
- Public APIs remain available and stable.

Dependencies

- Gmail API, Chrome Extension APIs, YouTube Data API.
- PostgreSQL, FAISS, Redis.
- Sentence-BERT models, Optional LLM services.

Design Details

The system integrates multiple platforms (Gmail, YouTube, Chrome), uses local processing and databases (PostgreSQL, FAISS, Redis), and works through secure APIs without modifying external services. It highlights that the design is local-first, portable, compatible with existing systems, and requires only minimal setup permissions.

The system depends on multiple platforms and technologies, including browser environments, external service APIs, local processing modules, and database systems. It integrates with platforms such as Gmail, YouTube, and Chrome through secure API and extension interfaces to collect user activity data. Internally, it relies on processing components for filtering, enrichment, and semantic embedding, along with storage systems like PostgreSQL, FAISS, and Redis for structured memory management, vector search, and caching. Only minimal configuration changes are required to these platforms, primarily permission authorization and API access setup, as the architecture is designed to operate locally without modifying external services. This ensures compatibility, portability, and secure integration while maintaining independence from third-party system changes.

Design Details(Characteristics and Innovation)

Design Characteristics

- **Interoperable:** Integrates heterogeneous sources into a unified memory system
- **Secure:** Local-first architecture; no typed text or private browsing captured
- **Performance:** <2s query retrieval via filtered semantic search
- **Maintainable:** Modular modules (capture, processing, retrieval, analytics)
- **Portable:** Runs on standard personal machines without cloud dependency

Innovation

- **Effort-aware retrieval** combining semantic similarity with engagement signals
- **Deterministic ranking** instead of opaque AI-only decisions
- **Intent-based admission filtering** preventing noisy or accidental data storage
- **Two-tier attachment processing** (lightweight preview + on-demand deep extraction)
- **Cross-source unified retrieval** across email, web, and video memory streams
- **Session-level behavioral modeling** for focus and fragmentation analysis.

Design Details(Quality Attributes)

Novelty

Echo introduces an intent-aware memory architecture that combines semantic similarity with engagement signals to determine relevance. Unlike traditional systems that rely only on keywords or embeddings, it models *what actually mattered to the user*, making retrieval cognitively meaningful rather than just technically accurate.

Legacy to Modernization

Echo bridges traditional data storage methods with modern semantic AI retrieval. It transforms static logs into intelligent, searchable memory representations.

Reusability

Core modules such as embedding generation, filtering, and ranking can be reused in other applications. This promotes extensibility for future research or product expansion.

Application Compatibility

The system integrates easily with existing applications through APIs and extensions. Its architecture avoids intrusive modifications, ensuring compatibility with current platforms.

Proposed System: Basic Approach & Results Obtained

Basic Approach

The system follows a **local-first, intent-aware memory architecture** that captures meaningful digital interactions across Gmail, Chrome, and YouTube, processes them into structured memory objects, and enables unified cross-source retrieval through semantic embeddings and deterministic ranking.

The approach consists of five stages:

1. Intent-based capture of user activity
2. Preprocessing and memory formation
3. Structured local storage
4. Semantic retrieval with effort-aware ranking
5. LLM-based synthesis of results

This ensures that only relevant, engagement-driven data is stored and retrieved.

Results Obtained

The proposed approach successfully demonstrates:

- Accurate filtering of irrelevant or accidental activity
- Unified search across heterogeneous sources
- Fast retrieval through vector similarity search
- Privacy-preserving local storage
- Meaningful ranking based on user engagement signals
- Explainable and deterministic retrieval behavior

The system effectively reconstructs contextual learning or work histories from fragmented digital activity.

Need for Changing the Approach

The initial design focused primarily on semantic relevance as the main retrieval signal. While this allowed accurate topic matching, it did not differentiate between content that was casually accessed and content that received sustained user attention.

This revealed a limitation: traditional semantic retrieval answers

“What content is related?”

but does not answer

“What content mattered to the user?”

User interaction patterns indicated that engagement intensity (time spent, revisits, interaction depth) is a strong indicator of importance. Ignoring this signal resulted in retrieval results that were technically relevant but not always cognitively meaningful.

Therefore, refinement of the retrieval strategy was necessary to better align the system’s behavior with human memory characteristics, where recall is influenced by effort and attention rather than similarity alone.

Updated Approach: Effort-Aware Retrieval



The initial design focused solely on semantic relevance. This answered "What content is related?" but not "What content mattered?" User engagement, measured by dwell time, revisits, and interaction intensity, proved to be a crucial indicator of importance.

New Approach: Prioritizing "Effort"

We enhanced retrieval by integrating an "Effort" signal, representing user engagement. The ranking now prioritizes results that are **Relevant AND highly engaged with**.

Benefits: More personalized, meaningful search; better alignment with human memory; reduced noise; improved prioritization of key materials.

Architecture Overview

The system is designed as a layered architecture that integrates **user interfaces**, **external data sources**, **processing modules**, **storage systems**, and **retrieval components** into a unified platform. The primary **logical user** interacts through the Search Interface and Dashboard UI, which allow querying stored memories, viewing insights, and managing settings. **External interfacing systems** such as Gmail API, YouTube API, and the browser extension provide activity data to the platform in a controlled and permission-based manner.

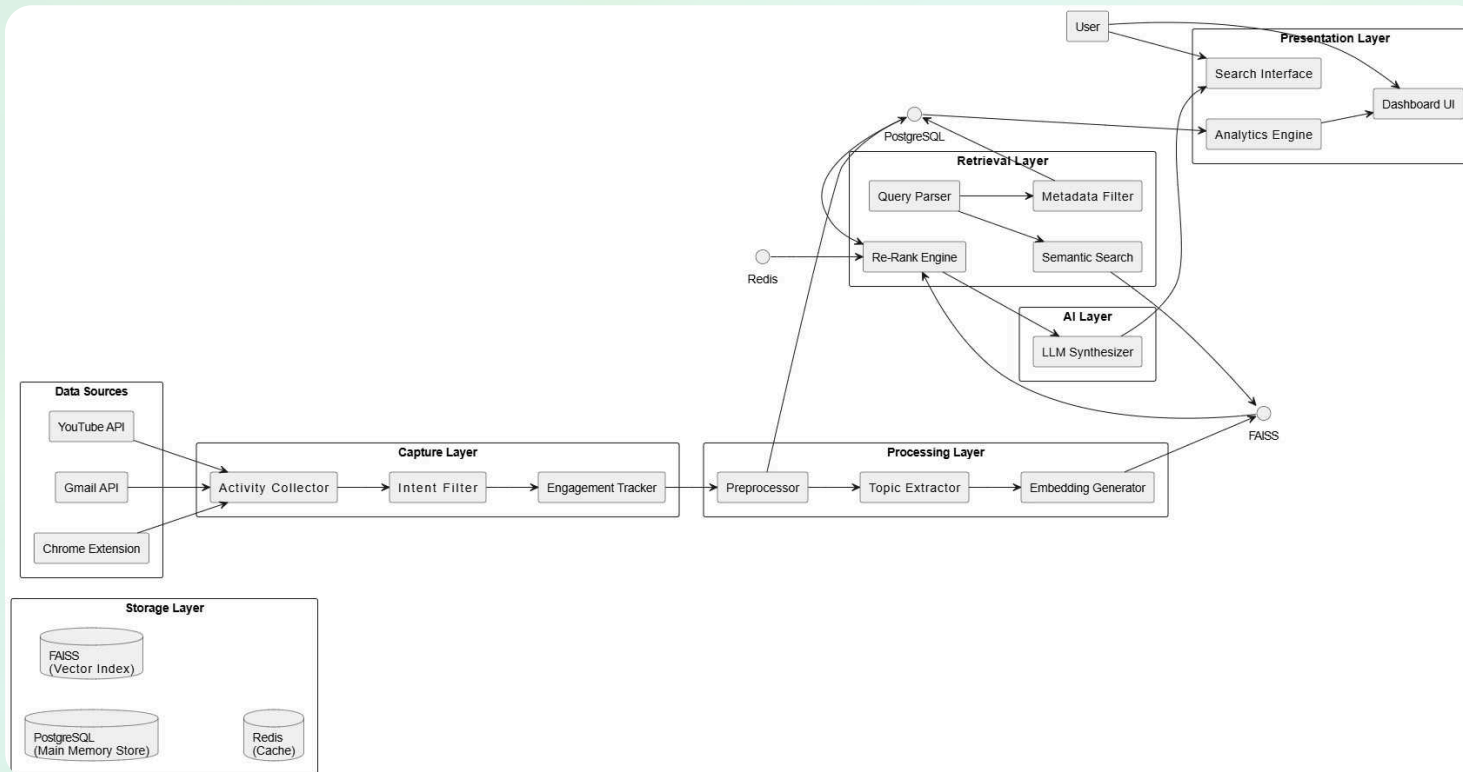
Internally, the application is divided into **application components** organized into functional layers. The Capture Layer collects and filters meaningful user interactions, the Processing Layer cleans and enriches data while generating semantic embeddings, and the Retrieval Layer handles query parsing, filtering, similarity search, and ranking. The AI Layer synthesizes retrieved results into readable responses, while the Presentation Layer displays outputs and analytics to the user.

The system relies on specialized **data components**: PostgreSQL stores permanent structured data, FAISS enables fast semantic search through vector indexing, and Redis supports temporary caching. These components collaborate through structured interaction, forming a pipeline workflow where captured data is processed, stored, retrieved upon request, and presented to the user. This layered collaboration ensures modularity, efficient retrieval, privacy preservation, and explainable system behavior.

Architecture

Layered Modular Monolithic Architecture.

High level Design Architecture Diagram



Layered
Architecture
Diagram of the
Echo System

Design Description

Design Description



 Modular architecture

 Local-first privacy model

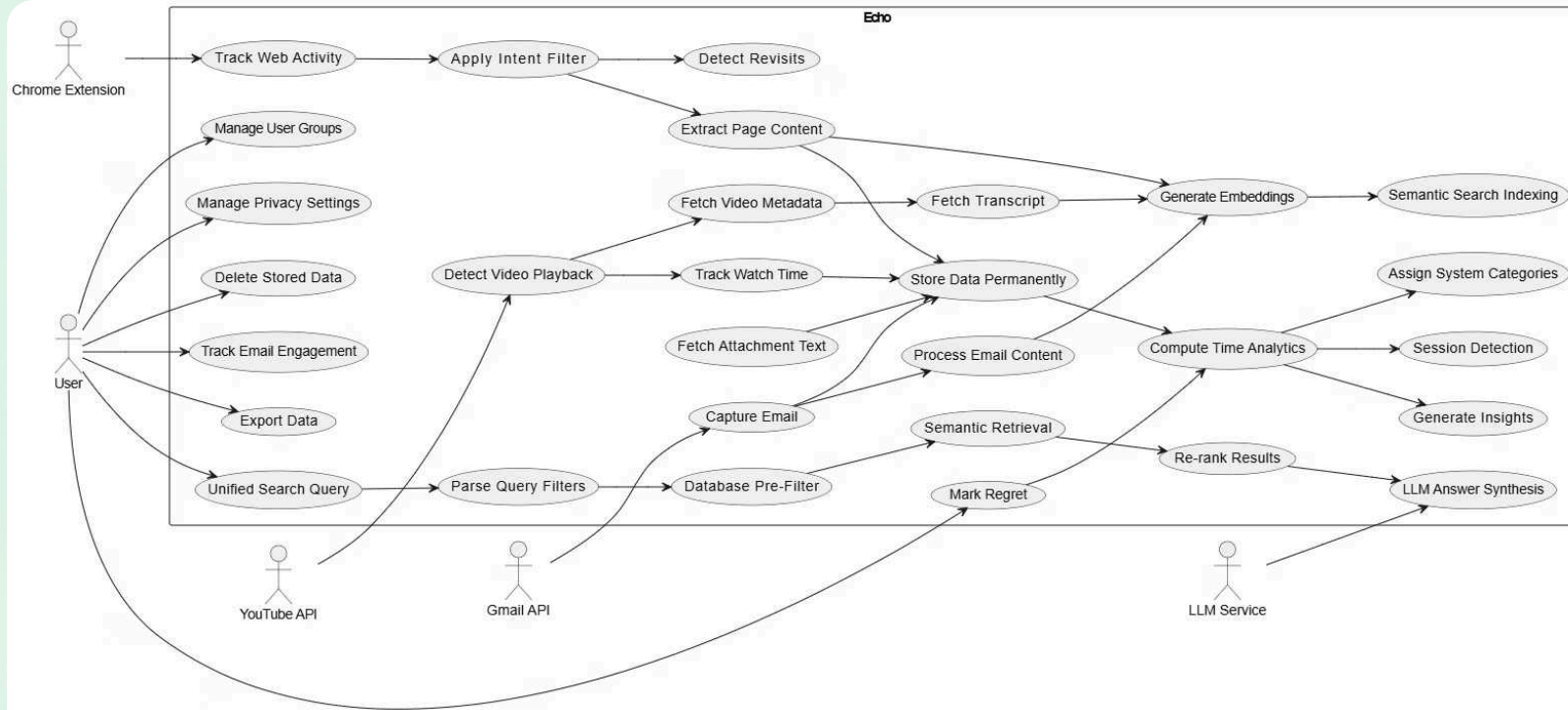
 Deterministic retrieval logic

 Local-first privacy model

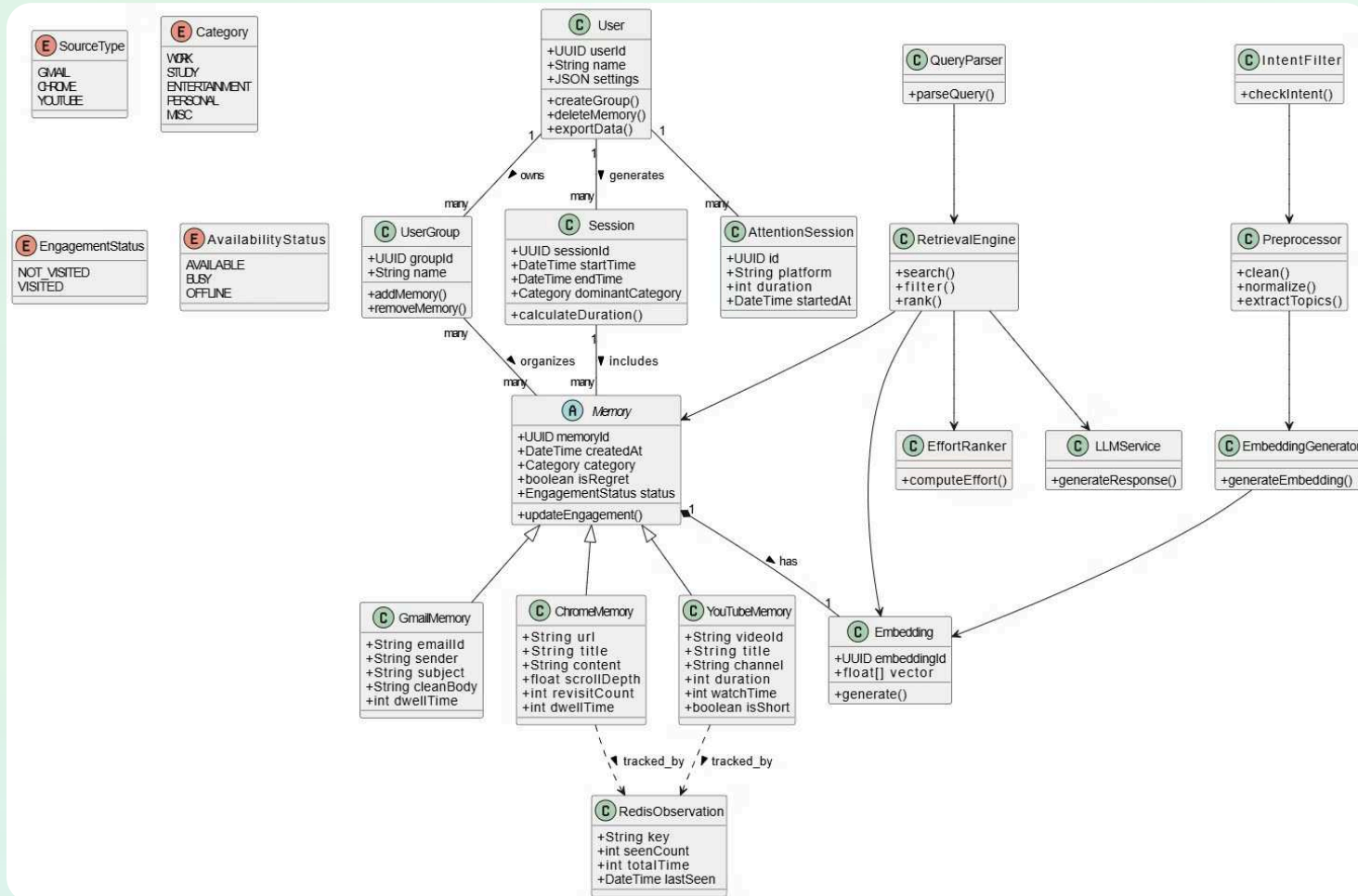
“ Key Design Principle:

Retrieval prioritizes what mattered to the user,
not just what matched the query.

- **Capture Layer** – Collects activity from Gmail API, Chrome Extension, and YouTube metadata.
- **Intent Filtering Layer** – Stores only engaged interactions using dwell time, scroll depth, and revisit signals.
- **Processing Layer** – Cleans content, extracts topics, and generates semantic embeddings.
- **Storage Layer** – PostgreSQL stores canonical data; FAISS indexes embeddings; Redis handles temporary state.
- **Retrieval Layer** – Deterministic pipeline combining filters + semantic similarity + engagement ranking.
- **Synthesis Layer** – LLM generates answers only from retrieved results.

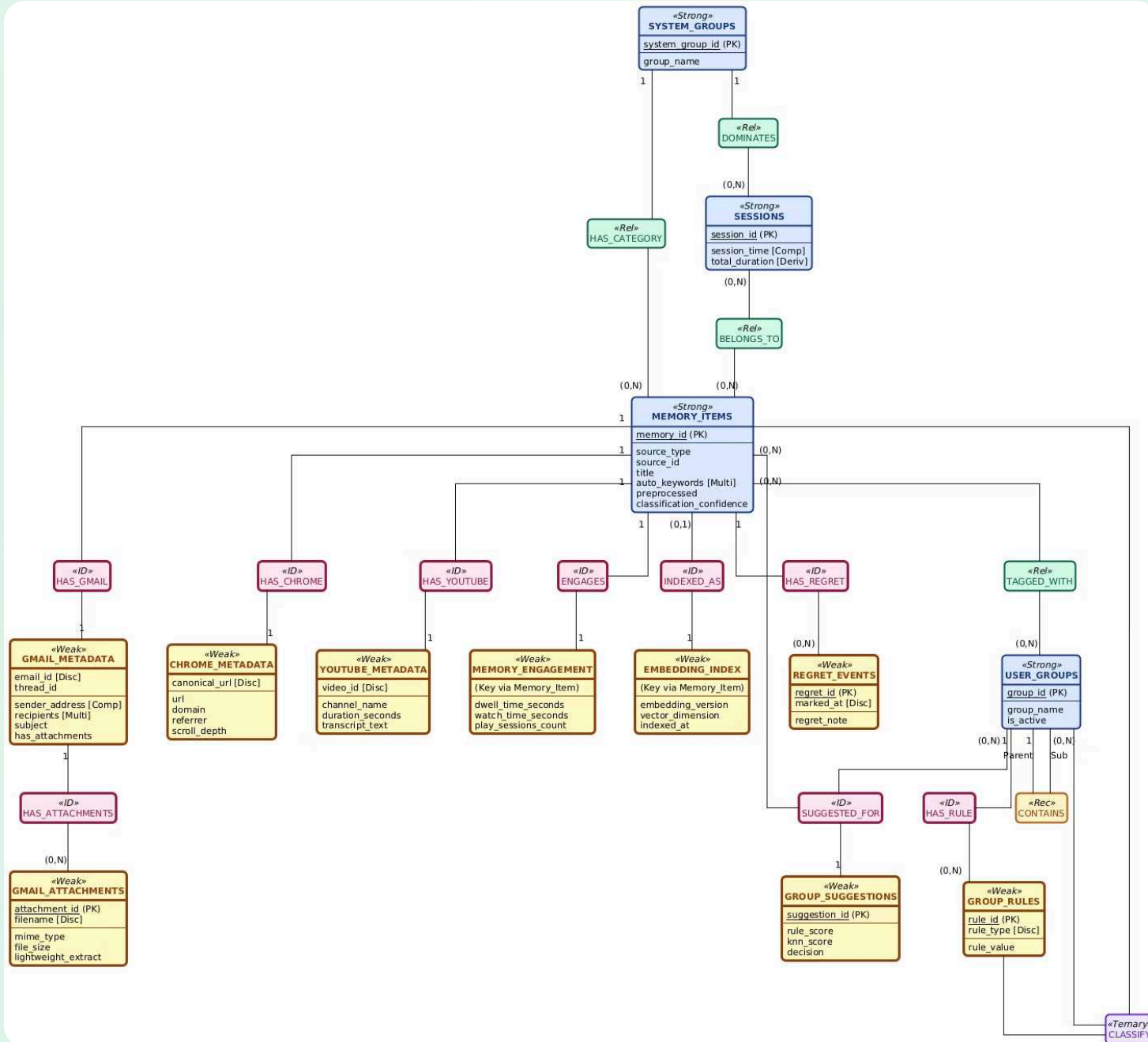


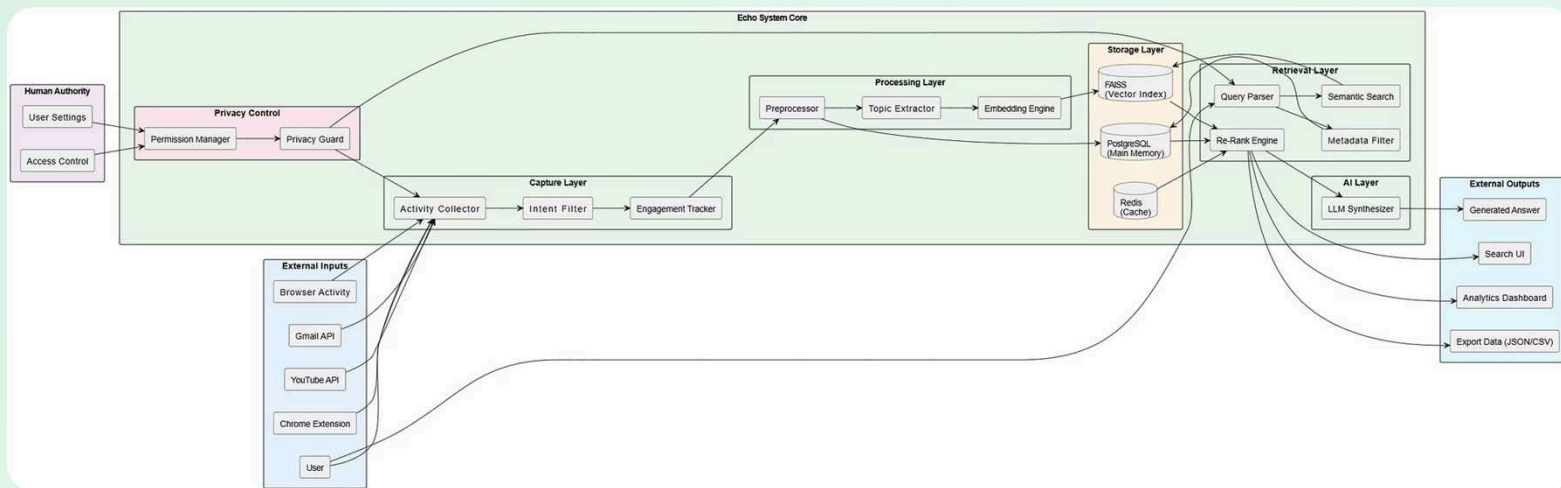
Use Case Diagram



Class Diagram

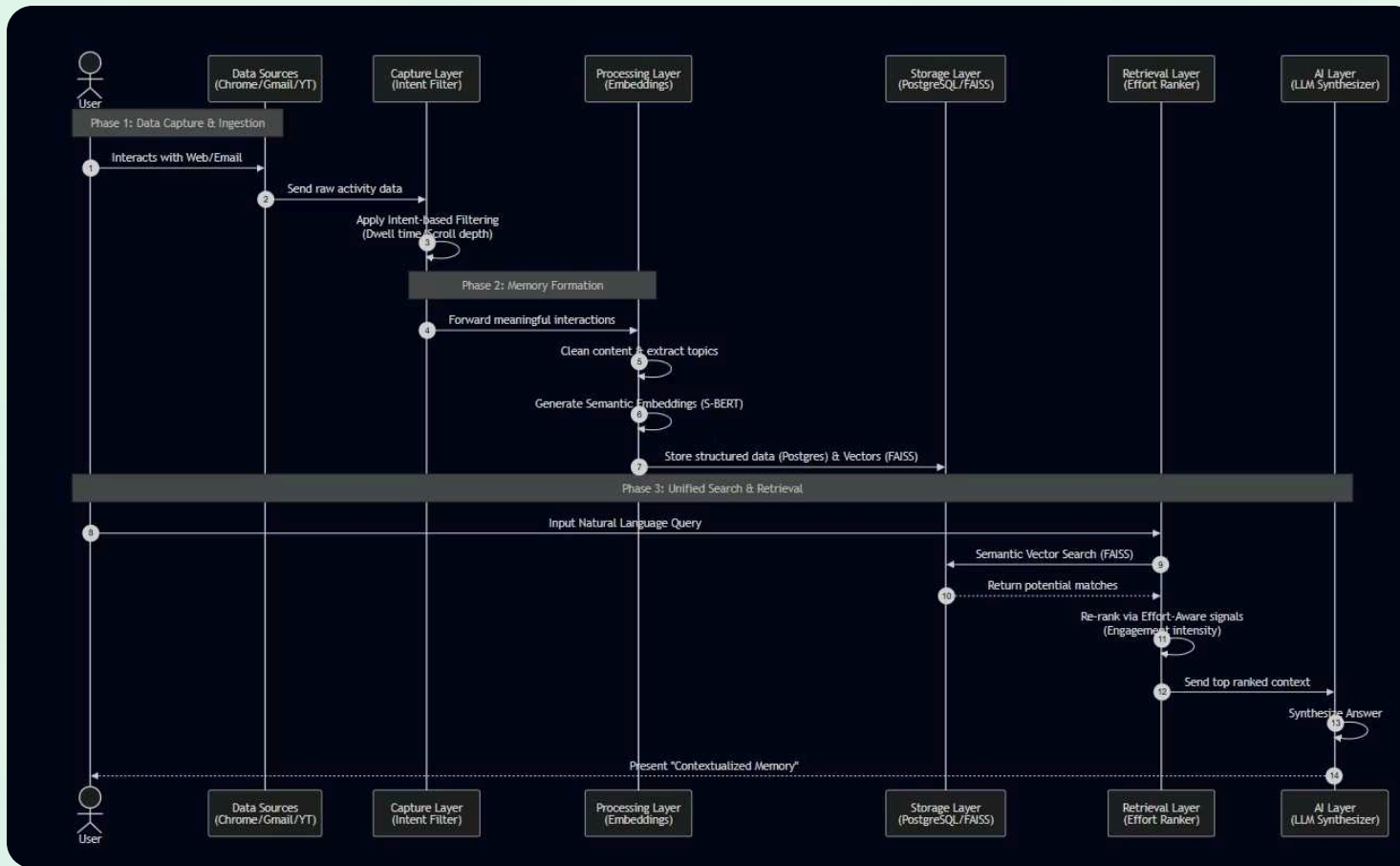
ER Diagram





External Interfaces

Sequence Diagram



Capstone (Phase-I ,Phase-II and Phase -III) Project Timeline

Phase I: Foundation (Aug 2025 – Dec 2025)

- Finalized and froze project scope
- Defined problem statement and objectives
- Conducted literature survey and technology gap analysis
- Explored potential data sources (Gmail, Chrome, YouTube)
- Completed Phase-I ESA documentation and review

Outcome: Concept validated and groundwork prepared for structured implementation in Phase II.

Phase II: HLD & Partial Implementation (Jan — April 2026)

Objective: Design system architecture and implement core unified memory functionality.

Current Status

- High-Level Design (HLD) completed
- System architecture defined
- Gmail data ingestion pipeline implemented
- Gmail data stored in PostgreSQL

Ongoing & Upcoming Work

- Complete Gmail preprocessing & retrieval
- Ensure Gmail ingestion reliability
- Design & implement Chrome ingestion
- Integrate Chrome data into unified storage

Phase II Target Outcome

- Gmail module fully implemented
- Chrome History integrated at intermediate level

Phase III: Advanced Implementation (April – December 2026)

Objective: Complete remaining modules and stabilize the full Echo system.

Planned Focus Areas:

- •Completion of YouTube integration and full cross-source unification
- •Strengthening semantic search and embedding-based retrieval
- •Development of Digital Wellbeing dashboard and analytics
- •Implementation of Regret system and reflective nudges
- •Performance optimization and long-term storage stabilization
- •Security validation and privacy reinforcement
- •Preparation of technical report and research documentation

Outcome:

Fully integrated unified memory system with cross-platform recall and operational wellbeing analytics, and formal submission of the E.C.H.O. framework.

Team Effort Allocation and Phase-wise Task Distribution

Phase II: Partial Implementation *(Gmail Completion + Chrome Integration)*

Arjun Sai N & Mohammed Shehzaad Khan

Complete Gmail preprocessing pipeline

Finalize ingestion reliability

Ensure retrieval compatibility for unified search

Validate full Gmail workflow end-to-end

Mrigank Jain & Mohammed Mir Fazlai Ali

Design and implement Chrome ingestion logic

Capture and normalize browsing metadata

Integrate Chrome into unified storage

Enable cross-source search with Gmail

Phase II Outcome:

Fully functional Gmail module and intermediate Chrome integration with operational unified search.

Phase III: Advanced Integration & Completion

(YouTube + Wellbeing + System Stabilization)

Arjun Sai N

- Implement and integrate YouTube ingestion module
- Ensure cross-source compatibility across all platforms

Mohammed Shehzaad Khan

- Refine semantic search and embedding-based retrieval
- Integrate and optimize LLM-based synthesis layer

Mrigank Jain

Oversee security validation and privacy hardening

Conduct system testing and performance optimization

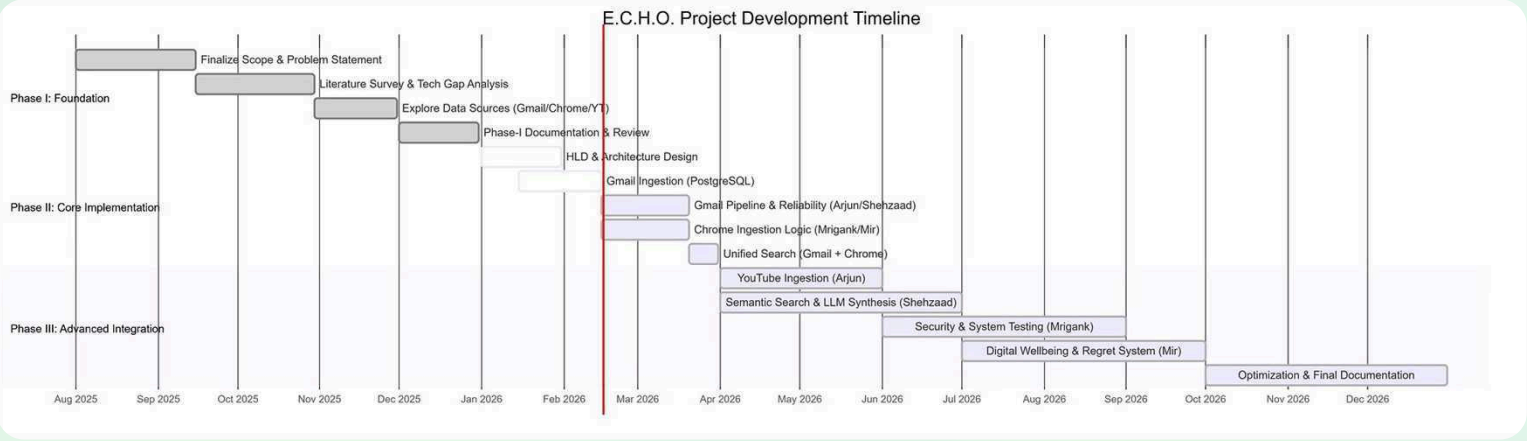
Mohammed Mir Fazlai Ali

Develop Digital Wellbeing dashboard

Implement regret system and behavioral analytics

Phase III Outcome:

Fully integrated multi-source memory system with stabilized performance and operational analytics



Project Timeline

The Future of Digital Interaction with Echo

Echo is built on an architecture designed to **transform personal memory at scale**, creating the ultimate intelligence layer for your digital existence.

This foundation unifies **everything**—every desktop app (Slack, GitHub, VS Code, Notion), every file, every conversation, every moment of learning. Through **agentic AI and MCP integration**, Echo becomes your **universal knowledge companion**

Echo isn't just tracking time—it's **capturing human intelligence**. While competitors offer screenshots or cloud-dependent solutions, Echo delivers **true ownership**: local-first, infinitely extensible, privacy-preserving.

The vision: Echo becomes the operating system for human memory—the layer between you and every digital tool you use, making your entire digital life searchable, analyzable, and truly yours.

Conclusion

ECHO represents a paradigm shift in how we interact with our digital memories.

Key Achievements:

- Privacy-first local architecture
- Intelligent, noise-reducing capture
- Effort-aware retrieval of important content
- Cross-source semantic search

Unlike cloud-based alternatives, ECHO empowers users with complete ownership and privacy over their digital memories through its modular, extensible design.

Looking Ahead: Phase III will integrate YouTube, launch digital wellbeing analytics, and deliver a complete memory system.

ECHO isn't just remembering your digital past—it's understanding it.

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